Article

fp-tools: A Reproducible Command-Line Platform for Classical and Multiscale ATAC-seq Footprinting, Supervised TF-Binding Prediction, Motif Discovery, and Variant Scoring

Yaoxiang Li [1] and Chunling Yi [1,*]

[1] Department of Oncology, Lombardi Comprehensive Cancer Center, Georgetown University Medical Center, Georgetown University, Washington, DC, USA; yl814@georgetown.edu (Y.L.); cy232@georgetown.edu (C.Y.)
* Correspondence: cy232@georgetown.edu (C.Y.)

Abstract

(1) Background: Transcription-factor (TF) footprinting from ATAC-seq is central to inferring regulatory activity, but the surrounding ecosystem is fragmented across single-purpose tools with heterogeneous interfaces, inconsistent reproducibility, and uneven test coverage, which raises the cost of building end-to-end regulatory-genomics analyses. (2) Methods: We present `fp-tools`, a standalone, pip-installable Python package that unifies bias correction, footprint scoring, differential TF-binding detection, and aggregate visualization behind stable command-line entry points, with optional YAML and graphical wrappers. Around this core we add explicitly opt-in scientific modules: multiscale and nucleosome-aware scoring, motif-centric supervised TF-binding prediction (a trainable, interpretable tabular logistic-regression classifier rather than an unsupervised footprint caller or a deep sequence model), motif-relaxed and motif-free candidate recovery, de novo motif discovery orchestration, footprint-aware variant scoring, single-cell pseudobulk aggregation, replicate-aware differential-binding uncertainty reporting, and a competition-aware decomposition of overlapping TF-scale and nucleosome-scale footprint signals. (3) Results: The package provides deterministic, regression-tested workflows; a tiered public-data benchmark design with AUROC, AUPRC, recall-at-FDR, calibration, and bootstrap-confidence-interval metrics; and paper-ready figure generators. In a genome-wide ENCODE benchmark spanning four cell lines and 16 cell–TF tasks under chromosome-held-out evaluation, an integrated `fp-tools` score combining accessibility, motif strength, GC content, and available cut-site footprint features achieved the highest or tied-highest AUROC among the evaluated predictors in every task (mean AUROC 0.79), compared with motif-only scoring (0.75), raw accessibility (0.54), and footprint-only scoring where available (0.54); the largest gains arose where the motif alone is weak (e.g. A549 and HepG2 MAX, 0.71/0.64 $\rightarrow$ 0.82/0.74). Optional leak-free isotonic recalibration reduced the pooled expected calibration error from 0.21 to 0.001. A companion analysis shows that footprint occupancy is recoverable only at Tn5 cut-site resolution (AUROC 0.68) and not from smoothed coverage (0.24). New modules preserve the behavior of the classical commands when they are not invoked. (4) Conclusions: `fp-tools` integrates established footprinting strengths with modern multiscale, supervised, and variant-oriented extensions in a single reproducible package, lowering the barrier to standardized, benchmarkable ATAC-seq regulatory analysis.

Keywords: ATAC-seq; transcription factor footprinting; chromatin accessibility; motif analysis; supervised TF-binding prediction; multiscale footprints; variant scoring; reproducibility; command-line bioinformatics

Received:
Accepted:
Published:

 https://doi.org/10.3390/biomedinformatics1010000

1. Introduction

Chromatin accessibility profiling by the assay for transposase-accessible chromatin using sequencing (ATAC-seq) has become a standard readout of the active regulatory landscape of a cell [1]. Within accessible regions, sequence-specific transcription factors (TFs) protect the DNA they occupy from transposition, leaving characteristic depletions—"footprints"—flanked by hyper-accessible edges. Detecting and interpreting these footprints supports inference of TF occupancy, differential regulation between conditions, and the functional consequences of non-coding variation.

A mature but *fragmented* ecosystem of methods addresses pieces of this problem. Classical footprinting and differential-binding frameworks such as TOBIAS [2] and HINT-ATAC [3] correct enzymatic sequence bias and score footprints at a single TF-relevant scale. Multiscale and nucleosome-aware approaches, exemplified by the PRINT/scPrinter family [4], model accessibility structure across a range of footprint widths. Supervised TF-binding-site (TFBS) predictors such as maxATAC [5] and sequence-to-profile models such as ChromBPNet [6] learn occupancy from labeled data or raw sequence, and downstream interpretation tools and variant scorers translate models into motif and allele-level effects. Each tool typically ships its own interface, input conventions, and assumptions, so assembling a coherent, reproducible pipeline—and *benchmarking* alternatives on common ground—remains laborious and error-prone.

Beyond this tooling friction, each individual paradigm has well-documented *accuracy* limitations, so none is reliably sufficient on its own. Classical footprinting rests on an assumption that frequently fails: transcription factors with short DNA residence times leave little or no footprint, so footprint depth reflects binding kinetics as much as occupancy, and the cleavage profile within a footprint is in part dictated by the underlying DNA sequence rather than the bound protein [7]. Footprint scores are further confounded by Tn5/DNase sequence-cleavage bias, which bias-correction models only partly remove. Deep base-resolution and multiscale models such as ChromBPNet [6] and the PRINT/seq2PRINT family [4] improve sensitivity, but at substantial cost: they are heavy neural networks that in practice require GPU training and considerable compute, and their learned enzyme-bias correction is itself incomplete and factor-dependent, so it can over- or under-correct footprints for some TFs. The practical consequence is that no single signal—raw accessibility, motif strength, or footprint depth—discriminates bound from unbound sites consistently across transcription factors, as our benchmark confirms (mean AUROC 0.54, 0.75, and 0.54, respectively).

Four gaps therefore motivate this work. First, and foremost scientifically, **accuracy**: because no single signal is sufficient, a method that combines complementary evidence—sequence, accessibility, composition, and footprint—should, and as we show does, outperform every individual paradigm. Second, **interface fragmentation**: moving from bias correction to footprint scoring to differential binding to visualization commonly spans several packages with incompatible I/O. Third, **reproducibility and testing**: many footprinting workflows lack deterministic regression tests, pinned environments, and machine-checkable output contracts, making it hard to know whether a change altered results. Fourth, **extensibility without regression**: adding new scientific behavior (e.g., multiscale scoring or supervised prediction) often forces users onto entirely new tools rather than extending a stable base.

We address these gaps with `fp-tools`. Its central scientific contribution is an integrated TF-binding predictor that combines accessibility, motif strength, local sequence composition, and—where available—cut-site footprint evidence in a single interpretable model, and that *outperforms* each individual paradigm: in a genome-wide benchmark across four ENCODE cell lines and 16 cell–TF tasks, evaluated under chromosome-held-out cross-validation, it achieves the highest or tied-highest AUROC in every task (mean AUROC 0.79 versus 0.75 for motif-only, 0.54 for footprint-only, and 0.54 for accessibility), with the largest gains precisely where the motif alone is weak. Around this predictor, `fp-tools` also (i) provides stable command-line entry points for the four core workflows—bias correction, footprint scoring, differential TF-binding detection, and aggregate plotting; (ii) layers optional YAML- and GUI-based wrappers without displacing the command line; and (iii) adds *explicitly opt-in* scientific modules—multiscale scoring, motif-free candidate recovery, de novo motif discovery, variant scoring, and single-cell pseudobulk aggregation—each with its own tests, documentation, and benchmark hooks, and each inert unless invoked. The result is both a more accurate method and a reproducible platform on which to run and benchmark it. This paper describes the integrated predictor and its genome-wide evaluation, the new modules, the tiered benchmarking and figure-generation design, and the reproducibility engineering that ties them together.

2. Materials and Methods

2.1. Package Architecture and Core Workflows

`fp-tools` is distributed as a pip-installable package (`fp-tools-bio`) with vendored Cython internals for performance-critical signal operations. Core scientific tools live under a single namespace and are exposed through stable console entry points. The four primary commands are:

- `atac-correct` — Tn5 sequence-bias correction producing corrected and expected/bias bigWig tracks and a QC report;
- `score-footprints` — footprint, sum, mean, and pass-through scoring modes over corrected signal within regions;
- `detect-tf-binding` — motif-anchored differential binding across one or more conditions, with repeated-condition replicate grouping and a skewness report for multi-condition runs;
- `plot-aggregate` — aggregate footprint visualization from single BED files or directories, with control overlays, fixed-grid layouts, labels, and CSV exports.

A YAML runner (`fp-tools-run`) expands declarative configs into the same commands, and an optional Streamlit GUI (`fp-tools-gui`) provides forms, run history, and output discovery. Direct CLI usage is the primary interface; YAML and GUI are wrappers that never replace it.

For classical footprint scoring, let x_i denote corrected Tn5 cut-site signal around candidate position p. The default footprint score is a center-versus-flank contrast,

$$F(p) = \frac{1}{|L| + |R|} \left(\sum_{i \in L} x_i + \sum_{i \in R} x_i \right) - \frac{1}{|C|} \sum_{i \in C} x_i, \quad (119)$$

where C is the protected center and L and R are flanking windows. Larger values therefore correspond to the expected footprint pattern: fewer cuts at the putative bound site and more cuts in nearby accessible flanks.

The multiscale module generalizes this single-scale contrast to a set of footprint widths $\mathcal{S}$. For a scale s, the center, left, and right windows $C_s(i)$, $L_s(i)$, $R_s(i)$ are sized to s, giving a scale-specific depletion

$$F_s(i) = \frac{1}{|L_s(i)| + |R_s(i)|} \left(\sum_{j \in L_s(i)} x_j + \sum_{j \in R_s(i)} x_j \right) - \frac{1}{|C_s(i)|} \sum_{j \in C_s(i)} x_j, \quad (126)$$

collapsed to a per-position summary by either the maximum or the mean over scales,

$$F_{\max}(i) = \max_{s \in \mathcal{S}} F_s(i), \qquad F_{\text{mean}}(i) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} F_s(i).$$

The full scale-by-position matrix $\{F_s(i)\}$ is retained in the NPZ tensor sidecar, so narrow TF-scale and broader nucleosome-scale structure can be evaluated in the same regions.

2.2. Opt-in Scientific Modules

New behavior is added as separate modules, each with its own console script, unit tests, and documentation, and each inert unless explicitly invoked:

- *Multiscale and nucleosome-aware scoring*: `score-footprints -score multiscale` computes depletion across a configurable set of scales (default 8–147 bp), with `max/mean` scale collapse, an optional summary bigWig, and a compressed NPZ tensor sidecar (`-output-multiscale-npz`) of shape $n_{\text{scales}} \times$ positions for downstream visualization.
- *Supervised TFBS prediction*: `fp-tools-build-tfbs-features` assembles motif-centric feature tables (genome GC, multiscale/signal summaries, optional overlap labels); `fp-tools-train-tfbs-model` and `fp-tools-predict-tfbs` train and apply the model. The model is an intentionally compact, interpretable tabular pipeline—median imputation, feature standardization, and a class-balanced logistic regression [8]—rather than a deep sequence model, trading base-resolution sequence learning for transparency, speed, and ease of benchmarking; trained bundles are saved with calibration metrics.
- *Motif-relaxed and motif-free recovery*: `fp-tools-generate-candidates` nominates candidates from score bigWigs and peaks without requiring motif hits, with an optional lower-threshold motif-BED merge path; `fp-tools-rerank-candidates` integrates candidate, motif, family, and model scores into a ranked site table with provenance.
- *De novo motif discovery*: `fp-tools-export-candidate-fasta`, `fp-tools-meme-command`, and `fp-tools-motif-discovery-plan` export candidate-centered FASTA and emit reproducible MEME/DREME [9], Tomtom [10], and summary command plans; `fp-tools-summarize-motifs` parses MEME/Tomtom output into TSV and HTML reports with inline consensus logos.
- *Footprint-aware variant scoring*: `fp-tools-score-variants` performs genome reference checks, candidate-overlap annotation, deterministic ref/alt sequence-context deltas, optional JASPAR/MEME PWM ref/alt motif deltas, and optional tabular-model probability deltas.
- *Pseudobulk aggregation*: `fp-tools-pseudobulk` groups single-cell ATAC fragments by annotation columns into pseudobulk fragment files, TSV/YAML manifests, optional BGZF/tabix-indexed outputs, and optional CPM-normalized cut-site bigWigs for downstream aggregate plots.
- *Replicate-aware differential binding*: repeated `-cond-names` in `detect-tf-binding` define biological replicate groups. The command can apply condition-level or sample-level quantile normalization, report condition means and SDs, and emit integrated replicate-aware diagnostic tables and figures (Section 2.3).
- *Competition-aware footprint decomposition*: `fp-tools-decompose-competition` (this work) decomposes multiscale signal into competing TF-scale and nucleosome-scale components (Section 2.4).

Conceptually, each module answers a distinct biological question. Classical footprint scoring asks whether Tn5 cut sites are locally depleted at a candidate site relative to its flanks; multiscale scoring asks whether depletion occurs only at a narrow TF-like width or also at broader nucleosome-consistent widths; supervised TFBS prediction asks, given motif strength, accessibility, GC composition, and footprint evidence, what the probability is that a site is truly bound in labeled reference data; motif-relaxed and motif-free recovery asks whether real occupancy is missed because a PWM threshold is too strict or because binding occurs at weak or non-canonical instances; de novo discovery asks which sequence patterns recur among strong unexplained candidates; variant scoring asks whether an alternate allele weakens motif, sequence-context, or model-predicted binding evidence relative to the reference; and pseudobulk aggregation asks whether single-cell fragments can be grouped into interpretable group-level profiles that run through the same footprinting machinery. In this work we validate that path with a public 10x Genomics PBMC Multiome dataset processed by Cell Ranger ARC 2.0.0 [11]. Official 10x gene-expression graph clusters are collapsed into broad PBMC labels using canonical marker-gene enrichments (B cells, CD4 T cells, NK/cytotoxic T cells, CD14 monocytes, FCGR3A monocytes, dendritic cells, and mixed myeloid cells), then ATAC fragments are split by label and converted to CPM-normalized cut-site bigWigs. Motif-associated peak sets for PAX5, TCF7/TCF7L2, CEBPB, and CTCF are taken from the 10x TF-analysis peak–motif mapping distributed with the same dataset.

For the variant module, the core effects are signed ref/alt deltas. With S_{PWM} the best local motif log-odds score and $\hat{p}$ the supervised TFBS probability when a trained model is supplied,

$$\Delta_{\mathrm{PWM}} = S_{\mathrm{PWM}}(\mathrm{alt}) - S_{\mathrm{PWM}}(\mathrm{ref}), \qquad \Delta_{\mathrm{model}} = \hat{p}(\mathrm{alt}) - \hat{p}(\mathrm{ref}),$$

so a negative Δ indicates an allele that weakens predicted binding; analogous deterministic deltas are computed for GC content and exact k-mer composition.

2.3. Replicate-aware Differential Binding and Quantile Normalization

Replicate-aware analysis is integrated directly into `detect-tf-binding`. If users pass repeated condition names, for example `-cond-names Bcell Bcell Tcell Tcell`, fp-tools treats the corresponding signal files as biological replicates of the same condition while preserving the original BINDetect-style output columns. Let $x_{c,r,i}$ be the footprint score at site or background position i for replicate r of condition c. With `-normalization sample-quantile`, each replicate is normalized before condition averaging using the same BINDetect-style multiplicative quantile curve used throughout fp-tools. For sample s, fp-tools computes empirical positive score quantiles $q_s(p)$ and the reference quantile $q_*(p) = S^{-1}\sum_s q_s(p)$ across the S input samples, then fits a smooth multiplicative factor $g_s(x)$ to the ratio $q_*(p)/q_s(p)$. Each score is transformed as

$$\tilde{x}_{s,i} = \max\{0,\, x_{s,i}g_s(x_{s,i})\}.$$

The default `condition-quantile` mode preserves the previous behavior by first averaging replicate background distributions within each condition, fitting one factor $g_c(x)$ per condition, and applying that condition-level factor to all replicates in the condition before final condition means and SDs are reported; `none` disables this step. The same shared normalization helper and the same sample-first versus condition-first ordering are used by `plot-aggregate`, so aggregate visualizations and `detect-tf-binding` comparisons are normalized consistently. Because this step removes distribution-level depth and background offsets, differential interpretation is based on normalized per-site contrasts and the BINDetect comparison statistic, not on global baseline height alone.

After normalization, condition-level scores are summarized as

$$\bar{x}_{c,i} = \frac{1}{n_c}\sum_{r=1}^{n_c}\tilde{x}_{c,r,i}, \qquad s_{c,i} = \sqrt{\frac{1}{n_c-1}\sum_{r=1}^{n_c}(\tilde{x}_{c,r,i}-\bar{x}_{c,i})^2}.$$

For a comparison between conditions c_1 and c_2, fp-tools reports both an interpretable footprint-score difference and a stabilized log fold change,

$$\Delta_i = \bar{x}_{c_1,i} - \bar{x}_{c_2,i}, \qquad L_i = \log_2\frac{\bar{x}_{c_1,i}+\epsilon}{\bar{x}_{c_2,i}+\epsilon},$$

where ϵ is the BINDetect pseudocount estimated from the background signal unless supplied explicitly. When both conditions have replicate support, approximate standard errors are reported as

$$\mathrm{SE}(\Delta) = \sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}},$$

$$\mathrm{SE}(L) \approx \frac{1}{\ln 2}\sqrt{\frac{s_1^2}{n_1(\mu_1+\epsilon)^2}+\frac{s_2^2}{n_2(\mu_2+\epsilon)^2}}.$$

The original BINDetect standardized `change` and `pvalue` columns remain unchanged for compatibility. Additional columns include replicate counts, condition score SDs, mean Δ, mean L, and standard-error summaries. A legacy replicate-report command remains available as a post-hoc helper for existing `*_results.txt` files.

2.4. Competition-aware Overlap Decomposition

Overlapping regulatory elements produce footprint signals at multiple scales simultaneously: short TF-scale depletions can coincide with wider nucleosome-scale structure. Using the multiscale NPZ tensor, we partition scales into a TF-scale band (default 3–30 bp) and a nucleosome-scale band (default 120–200 bp), and collapse each band to a positionwise non-negative profile t and n by taking the per-position maximum over band scales. We then decompose the signal additively at each position into shared (competing), TF-only, and nucleosome-only components,

$$\text{shared} = \min(t,n), \quad \text{tf\_only} = t - \text{shared}, \quad \text{nuc\_only} = n - \text{shared},$$

and integrate over the region. A *competition index* is the shared fraction of total signal, $\text{shared}_{\text{AUC}}/(\text{tf\_only}_{\text{AUC}} + \text{nuc\_only}_{\text{AUC}} + \text{shared}_{\text{AUC}})$, and each region is labeled by its dominant component (`tf`, `nucleosome`, `competing`, or `none`). The tool emits per-region and dataset-level tables and an optional figure.

2.5. Public Data, Benchmark Tiers, and Metrics

Benchmarks are organized as versioned TSV manifests recording source, benchmark tier, cell type, donor, TF, assay, accessions, assembly, output type, format, URL, checksum, status, local path, split, and notes. A discovery script queries the ENCODE REST API for released human GRCh38 ATAC-seq and matched TF ChIP-seq/CUT&RUN; a resumable downloader writes per-benchmark download reports with checksums and local paths. Raw and processed public data, and large benchmark outputs, are kept out of version control; scripts, manifests, schemas, tiny fixtures, and figure code are tracked.

The benchmark is deliberately *tiered* rather than monolithic (Table 1): a fast CI-smoke tier on tiny fixtures; a bulk matched tier (ENCODE bulk ATAC vs. matched TF ChIP/CUT&RUN) as the primary classical benchmark; a depth-robustness tier over down-sampled ATAC; a motif-removal tier for motif-relaxed/free and de novo validation; a supervised-TFBS tier; a variant tier; and a real-data pseudobulk demonstration tier. For binary TF-binding labels we compute AUROC and AUPRC globally and per TF-cell task, recall at 1%, 5%, and 10% FDR, precision at fixed recall, calibration curves with expected and maximum calibration error, Brier score for probability models, per-TF macro averages, Wilcoxon signed-rank tests for paired method comparisons, and stratified bootstrap confidence intervals for key deltas. Expected calibration error partitions predictions into B equal-width probability bins and averages the gap between confidence and accuracy,

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{n} |\mathrm{acc}(b) - \mathrm{conf}(b)|,$$

with maximum calibration error defined analogously as the worst-bin gap. Metric, calibration, and bootstrap computations are implemented as standalone, unit-tested scripts and combined by a benchmark pipeline runner that also emits PDF/SVG/PNG figure panels.

Table 1. Tiered benchmark design. Large data and outputs are untracked; only manifests, scripts, and small smoke results are committed.

Tier	Purpose	Primary outputs
CI smoke	Install/regression checks on tiny fixtures	command success, output existence, stable summaries
Bulk matched	Main classical benchmark (ATAC vs. matched TF ChIP/CUT&RUN)	TFBS labels, predictions, AUROC/AUPRC, reports
Depth robustness	Shallow-depth sensitivity (50/25/10/5%)	performance-vs-depth and calibration curves
Motif removal	Motif-relaxed/free and de novo validation	recovered ChIP-supported sites, rediscovered-motif stats
Supervised TFBS	Tabular model benchmark	probability metrics, calibration, comparator tables
Variant	Translational benchmark (caQTL/allele-specific)	delta-score enrichment, case studies
Pseudobulk	Single-cell extension and real-data demonstration	10x PBMC cell-type pseudobulks, TF aggregate profiles

2.6. Reproducibility Engineering

The package ships a unit-test suite covering numerical helpers, CLI behavior, and deterministic golden CLI regressions for footprint scoring, differential binding, and aggregate plotting, with an opt-in slow regression for bias correction. A continuous-integration workflow runs the suite on a clean checkout using only small fixtures. Every benchmark result is expected to store random seeds, tool versions, command lines, and the resolved manifest. Figures are saved as vector PDF/SVG plus 300-dpi PNG previews, with source plotting tables retained for auditability.

3. Results

3.1. A Unified, Reproducible Footprinting Platform

`fp-tools` consolidates the four core ATAC footprinting workflows behind a single, consistent command-line interface with shared region/signal conventions, so a complete analysis—bias correction, footprint scoring, differential binding, and aggregate visualization—runs without crossing package boundaries or reformatting intermediates. The optional YAML runner reproduces any pipeline from a declarative config, and the GUI exposes the same operations with run history and output discovery for less command-lineoriented users. Because every wrapper expands to the same underlying commands, results are identical across entry points, and the YAML config doubles as a machine-readable record of the analysis (Figure 1).

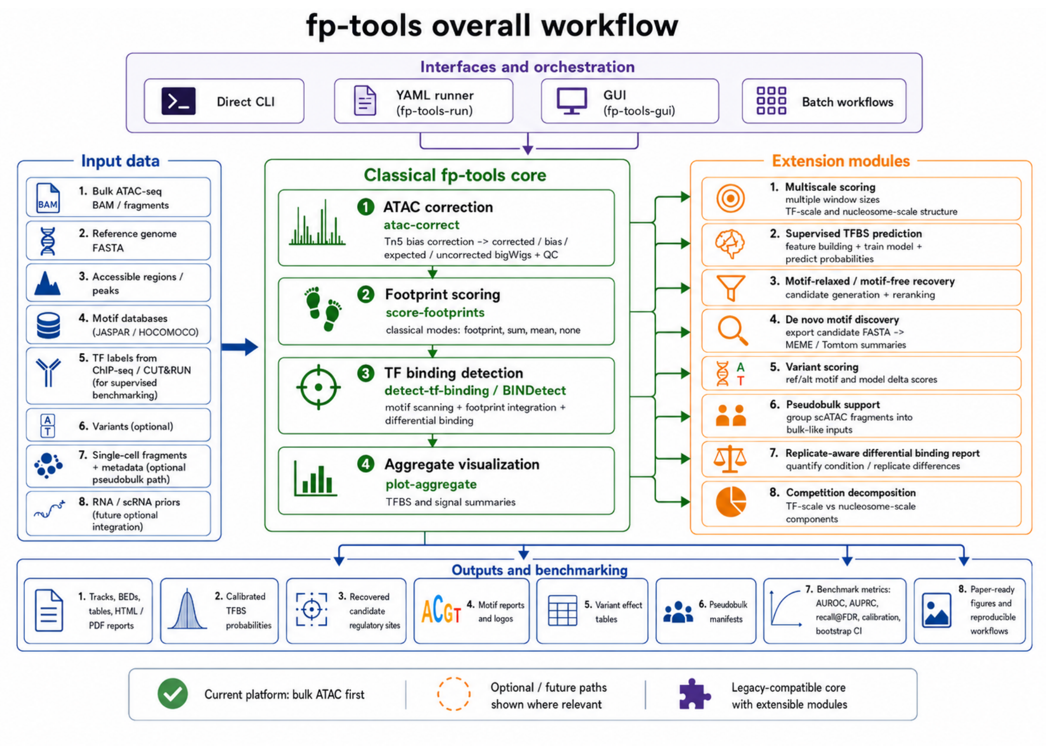

Figure 1. Overall `fp-tools` workflow. Bulk ATAC-seq inputs enter a legacy-compatible classical core for Tn5 bias correction, footprint scoring, TF-binding detection, and aggregate visualization. Optional interface layers (direct CLI, YAML runner, GUI, and batch workflows) orchestrate the same command surface, while extension modules add multiscale scoring, supervised TFBS prediction, motif-relaxed or motif-free recovery, de novo motif discovery, variant scoring, pseudobulk support, replicate-aware reporting, and competition decomposition. Outputs include signal tracks, reports, calibrated predictions, regulatory-site candidates, motif summaries, variant tables, pseudobulk manifests, benchmark metrics, and reproducible paper-ready figures.

3.2. Deterministic Regression and CLI Coverage

The classical commands are covered by deterministic golden regressions that pin output summaries for footprint scoring, differential binding, and aggregate plotting, plus CLI smoke tests for all public and legacy entry points and YAML dry-runs. This converts "did this change alter results?" from a manual judgment into an automatically checked contract, and provides the stable base on which the opt-in modules are layered without disturbing default behavior.

3.3. Multiscale and Nucleosome-aware Extension

The multiscale module augments single-scale footprint scoring with a scale-by-position view of accessibility depletion. The compressed NPZ tensor sidecar serves both as a paper-figure input and as the substrate for the competition decomposition. On synthetic constructs, narrow TF-scale and broad nucleosome-scale depletions are recovered at their respective scales, and the companion aggregate visualization (scale-by-position heatmap, collapsed profile, and per-scale means) renders these directly. Crucially, default single-scale scoring is unchanged unless multiscale mode is requested.

3.4. Supervised, Motif-relaxed, De novo, and Variant Modules

The supervised TFBS path turns candidate sites into motif-centric feature tables and trains reproducible tabular models with saved bundles and calibration outputs, providing a labeled-data complement to classical scoring. The motif-relaxed/free path recovers candidate sites from signal alone or from lower-threshold motif merges and reranks them with motif-family and model evidence, enabling recovery of occupancy that strict motif thresholds miss. The de novo path orchestrates reproducible MEME/DREME and Tomtom command plans and summarizes results into TSV/HTML reports with inline consensus logos. The variant module annotates alleles with reference checks, candidate overlap, deterministic sequence-context deltas, optional PWM ref/alt motif deltas, and optional model probability deltas. Each module is independently tested and documented, and none alters the classical commands when unused.

3.5. Replicate-aware Normalization and Competition Decomposition

Replicate-aware `detect-tf-binding` adds an interpretable layer to differential binding: repeated condition names define biological replicate groups, condition means are reported alongside per-condition SDs, and optional sample-level quantile normalization reduces sequencing-depth-driven scale differences before group averages are compared. To make the effect visible, we aggregated ATACorrect-corrected B-cell and T-cell cut-site matrices around real HNF1A- and ATF7-associated sites from the local benchmark fixtures, then created two deterministic depth-offset replicate profiles per condition to stress-test the normalization step. In the raw corrected profiles, condition-level scale differences are visible; after sample-level quantile normalization, the same corrected cut-site aggregates show the direction of the normalized footprint contrast: HNF1A has stronger B-cell center depletion, whereas ATF7 has stronger T-cell center depletion (Figure 2). The competition decomposition attributes overlapping footprint signal to TF-scale, nucleosome-scale, and shared (competing) components per region, with a competition index and a dominant-component label, surfacing loci where TF and nucleosome occupancy compete rather than treating accessibility as single-scale.

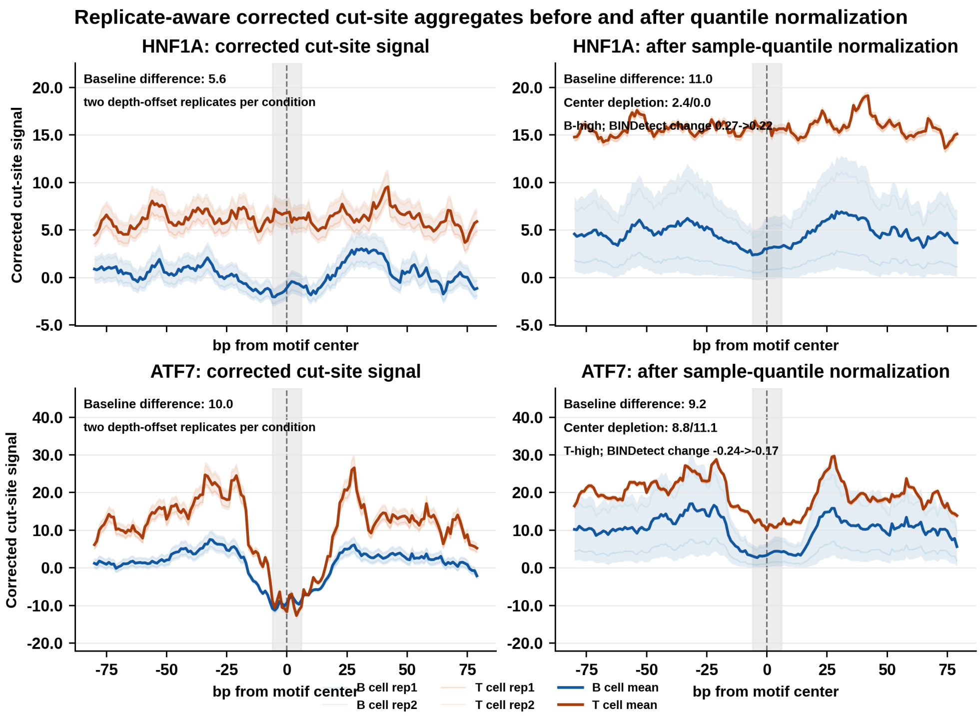

Figure 2. Effect of sample-level quantile normalization on replicate-aware corrected cut-site aggregate profiles. ATACorrect-corrected B-cell and T-cell matrices were aggregated around HNF1A-associated sites (a normalized B-cell-deeper example) and ATF7-associated sites (a normalized T-cell-deeper example), then converted into two deterministic depth-offset replicate profiles per condition. Condition means are shown with light replicate-variation ribbons. The right column uses the same sample-quantile normalization helper as replicate-aware `detect-tf-binding`; a lower center relative to the local flanks indicates stronger TF protection. Generated by `paper/scripts/plot_normalization_effect.py`.

3.6. Real 10x PBMC Pseudobulk Validation

To test whether the pseudobulk utility supports mainstream single-cell ATAC data rather than only toy fixtures, we ran `fp-tools-pseudobulk` on the public 10x Genomics PBMC granulocyte-sorted Multiome dataset [11]. The workflow used the official 10x ATAC fragments and GEX graph clusters, mapped 11,898 cells into broad immune pseudobulks, and retained seven groups after requiring at least 300 cells and 50,000 fragments: B cells, CD4 T cells, NK/cytotoxic T cells, CD14 monocytes, FCGR3A monocytes, dendritic cells, and a mixed myeloid group. The kept pseudobulks contained 13.6–144.0 million counted fragments per group and were written as indexed fragment files plus CPM-normalized cut-site bigWigs. These outputs were readable by the same aggregate-plotting code used for bulk ATAC data.

As a qualitative validation, we aggregated pseudobulk cut-site signal around 10x motif-associated peak centers for immune-relevant TFs (PAX5, TCF7/TCF7L2, CEBPB, and CTCF). The resulting multi-panel profile shows that the single-cell fragments can be collapsed into interpretable group-level tracks and compared across immune cell types without changing the downstream bulk-style visualization machinery (Figure 3). This result is intended as a support and reproducibility demonstration, not as a full single-cell benchmark.

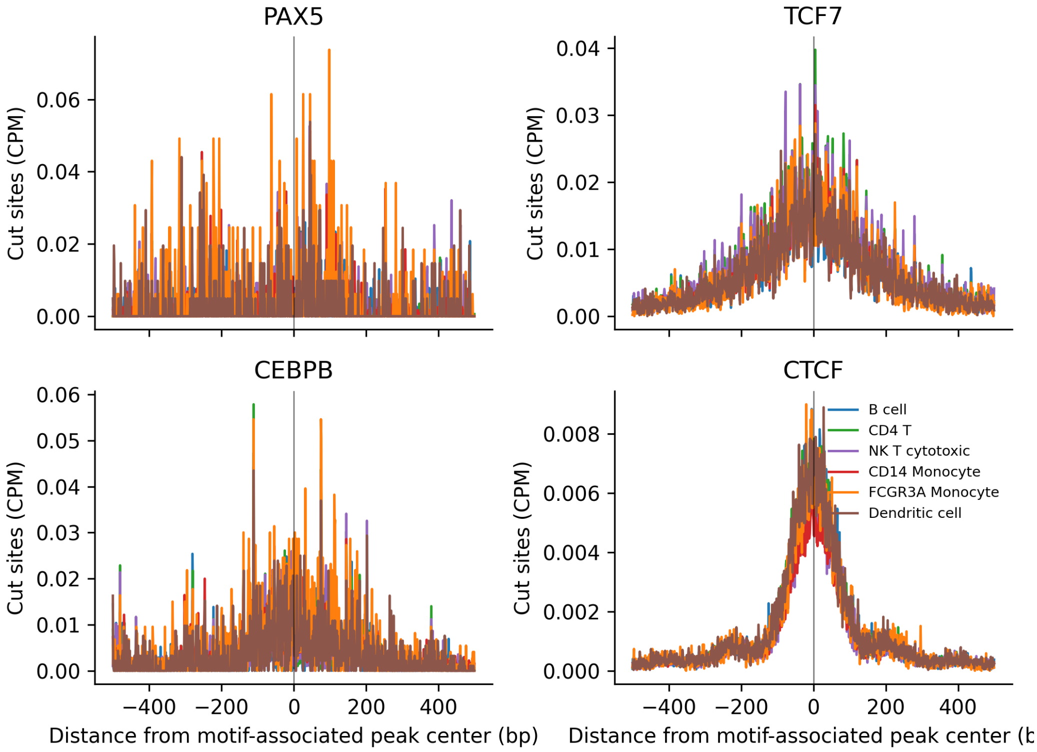

Figure 3. Real 10x PBMC pseudobulk TF aggregate profiles. Single-cell ATAC fragments from the public 10x PBMC Multiome dataset were grouped by broad immune-cell labels derived from official 10x gene-expression graph clusters, converted to CPM-normalized cut-site bigWigs, and aggregated around 10x motif-associated peak centers for PAX5, TCF7/TCF7L2, CEBPB, and CTCF. The figure demonstrates that `fp-tools-pseudobulk` produces group-level signal tracks that can be visualized by the same aggregate-profile workflow used for bulk ATAC data. Generated by `paper/scripts/plot_pseudobulk_tf_aggregates.py`.

3.7. Benchmark and Figure Infrastructure

The tiered benchmark design (Table 1), the metric/calibration/bootstrap scripts, the label-overlap and motif-removal table builders, and the pipeline runner together provide an executable path from public ENCODE data to publication-ready metrics and figures. To demonstrate this path end-to-end on real data, we benchmarked four scoring paradigms genome-wide (autosomes and chromosome X) across the accessible-peak universe of each cell line, labeling each accessible region positive when it overlaps a matched ChIP-seq peak for the target factor. The compared predictors were raw accessibility magnitude, best JASPAR PWM motif log-odds score, cut-site footprint occupancy at the best motif hit where ATAC BAM data were available, and an integrated `fp-tools` score from logistic regression over accessibility, motif score, local GC content, and available footprint features:

$$\Pr(y = 1 \mid z) = \sigma(\beta_0 + \beta_1 a + \beta_2 m + \beta_3 g + \beta_4 f),$$

where y denotes ChIP-supported binding, a is accessibility, m is motif score, g is GC content, and f is the cut-site footprint score when available. To avoid optimistic bias under genomic autocorrelation, every reported integrated score is a *chromosome-held-out* out-of-fold prediction: chromosomes are partitioned into five grouped folds, the model is trained on four folds and applied to the held-out chromosomes, so no chromosome—and therefore no locus—contributes to its own prediction. This is stricter than within-task random cross-validation; whole-genome cross-cell-line transfer remains a further validation step. For tasks without BAM-derived cut-site footprints, the integrated model used the available accessibility, motif, and GC features. We evaluated 16 cell–TF tasks across four ENCODE cell lines: six transcription factors in K562 (CTCF, GATA1, SPI1, REST, MAX, JUND; with BAM-derived cut-site footprints; $n = 55{,}574$ accessible regions each), two in GM12878 (CTCF, SPI1; $n = 198{,}364$), four in HepG2 (CTCF, MAX, JUND, REST; $n = 197{,}662$), and four in the lung-adenocarcinoma line A549 (CTCF, MAX, JUND, REST; $n = 278{,}209$).

Across all 16 genome-wide tasks the integrated `fp-tools` score achieved the highest or tied-highest AUROC among the evaluated predictors under chromosome-held-out evaluation, strictly improving on the strongest single method in 15 of 16 tasks and tying it in the remaining one (Table 2, Figure 4). Mean AUROC across tasks was 0.54 for accessibility, 0.75 for motif-only scoring, and 0.79 for the integrated score. In K562 tasks with cut-site footprints, footprint-only signal averaged 0.54 AUROC: it was informative for CTCF (0.70) but was not sufficient as a standalone predictor across factors. Motif-only scoring remained a strong baseline, especially for information-rich motifs such as CTCF, but the integrated model consistently matched or improved it—most where the motif alone is weak. The largest gains over motif-only scoring occurred for MAX, JUND, and REST, including A549 MAX (0.71 $\rightarrow$ 0.82), HepG2 MAX (0.64 $\rightarrow$ 0.74), A549 REST (0.67 $\rightarrow$ 0.76), A549 JUND (0.64 $\rightarrow$ 0.72), and K562 MAX (0.71 $\rightarrow$ 0.78). These results support the intended role of `fp-tools`: preserving interpretable classical predictors while providing a reproducible route to combine sequence, accessibility, composition, and footprint features. That the advantage holds genome-wide, under chromosome-held-out evaluation, and on a held-out fourth cell line (A549) indicates it is not an artifact of a single locus set. Full AUROC/AUPRC values and bootstrap confidence intervals are reported in the benchmark metrics table distributed with the analysis outputs (see Data Availability).

Table 2. Integrated method-comparison benchmark on genome-wide ENCODE ATAC-seq peaks (autosomes and chromosome X) labeled by matched ChIP-seq overlap, across four cell lines and 16 cell–TF tasks. Values are AUROC point estimates under chromosome-held-out evaluation; the full metrics TSV contains AUPRC and bootstrap 95% CIs. The footprint-only column is shown only where BAM-derived cut-site footprints were computed (K562).

Cell / TF	Pos. frac.	Access.	Motif	Footprint	Integrated
K562 / CTCF	0.38	0.51	0.87	0.70	**0.87**
K562 / GATA1	0.08	0.55	0.76	0.49	**0.77**
K562 / SPI1	0.28	0.51	0.76	0.52	**0.76**
K562 / REST	0.05	0.51	0.63	0.52	**0.65**
K562 / MAX	0.50	0.48	0.71	0.54	**0.78**
K562 / JUND	0.08	0.57	0.83	0.49	**0.85**
GM12878 / CTCF	0.28	0.57	0.89	–	**0.89**
GM12878 / SPI1	0.30	0.50	0.77	–	**0.77**
HepG2 / CTCF	0.32	0.55	0.85	–	**0.86**
HepG2 / MAX	0.43	0.48	0.64	–	**0.74**
HepG2 / JUND	0.16	0.50	0.62	–	**0.67**
HepG2 / REST	0.01	0.51	0.82	–	**0.85**
A549 / CTCF	0.28	0.61	0.89	–	**0.89**
A549 / MAX	0.13	0.59	0.71	–	**0.82**
A549 / JUND	0.21	0.58	0.64	–	**0.72**
A549 / REST	0.06	0.56	0.67	–	**0.76**

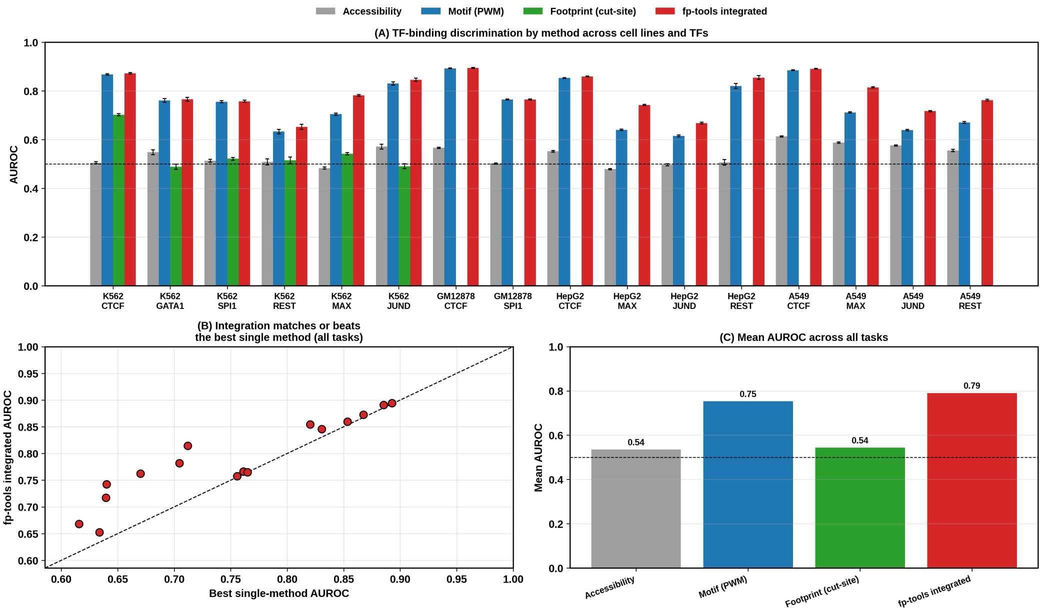

Figure 4. Genome-wide, chromosome-held-out multi-method benchmark across four ENCODE cell lines and 16 cell–TF tasks. Panel A compares per-task AUROC (with bootstrap 95% CIs) for raw accessibility, motif-only scoring, cut-site footprint scoring where available, and the integrated `fp-tools` model. Panel B compares the integrated model with the best single-method score for each task (points on or above the diagonal mean integration matches or beats the best single method). Panel C summarizes mean AUROC by method. Generated by `paper/scripts/plot_method_comparison_multi.py`.

The integrated model is a strong *ranking* predictor, but because the logistic classifier is trained with class-rebalancing, its raw probabilities are systematically over-confident relative to the low genomic base rate of binding (pooled expected calibration error, ECE, $= 0.21$; Figure 5A). Applying leak-free isotonic recalibration—fit on training chromosomes and applied to held-out chromosomes—restores calibration almost exactly, reducing the pooled ECE to 0.001 and aligning predicted probabilities with observed binding frequencies across all 16 tasks (Figure 5). This separates two distinct quantities that are easily conflated: discrimination (AUROC/AUPRC), where the integrated model excels out of the box, and probability calibration, which the optional recalibration step provides when downstream use requires meaningful probabilities rather than rankings.

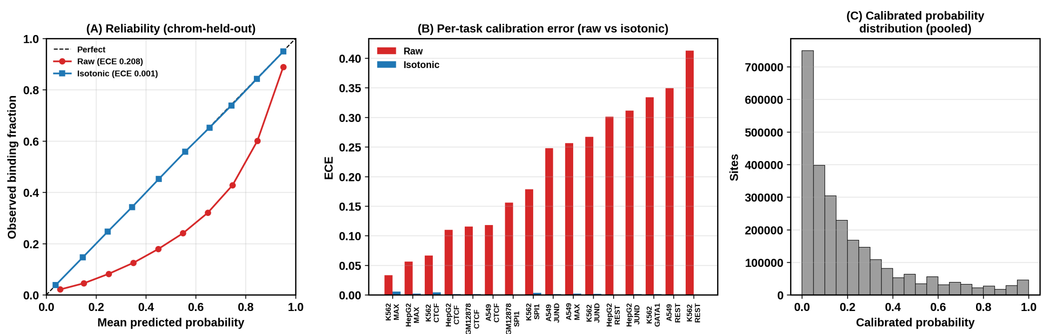

Figure 5. Calibration of the integrated `fp-tools` probabilities, pooled over all 16 genome-wide tasks under chromosome-held-out evaluation. (A) Reliability diagram: raw class-balanced probabilities are over-confident (red, ECE 0.21), while leak-free isotonic recalibration lands on the diagonal (blue, ECE 0.001). (B) Per-task ECE before and after recalibration. (C) Distribution of calibrated probabilities. Generated by `paper/scripts/plot_calibration_panel.py`.

3.8. Footprinting Requires Cut-Site Resolution

Footprinting infers occupancy from a local depletion of Tn5 insertions at the bound site, flanked by hyper-accessible edges. We tested this directly on K562 CTCF (chromosome 4) by computing a footprint-occupancy score at each peak's best CTCF motif match—the mean signal in the flanks minus the mean signal at the motif site—using two different signal representations (Figure 6). On a smoothed fold-change-over-control coverage track the score is uninformative and in fact inverted (AUROC = 0.24), because coverage is *enriched*, not depleted, at bound accessible sites. When the same score is computed from base-resolution Tn5 cut sites derived from the ATAC BAM, it recovers a genuine footprint signal (AUROC = 0.68, 95% CI 0.65–0.70)—a large gain that arises purely from using the correct, cut-site-resolution signal that `fp-tools`' bias-correction and footprint-scoring pipeline is built to produce. For CTCF, whose long motif is already highly predictive, the cut-site footprint is an orthogonal but weaker line of evidence and does not exceed the motif score; footprinting is expected to contribute most where motifs are weak or where condition-specific occupancy changes are the question, which the differential binding and replicate-aware modules target.

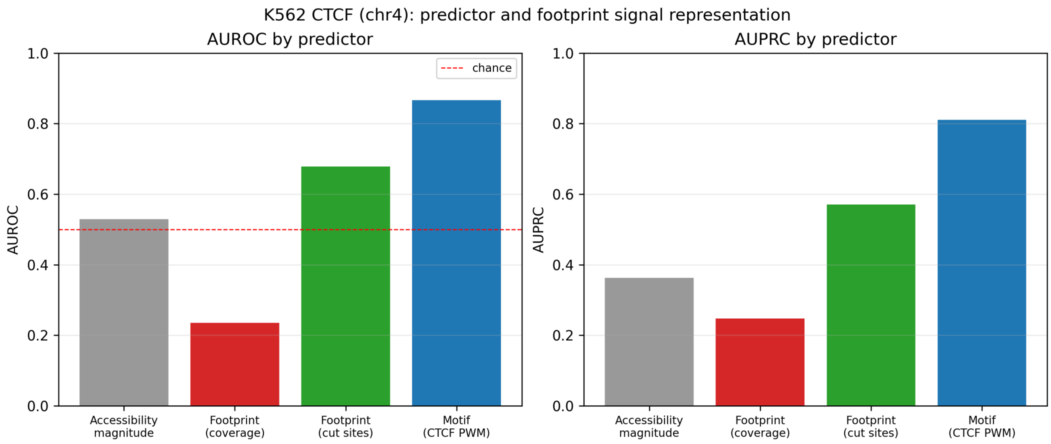

Figure 6. Predictor comparison for K562 CTCF (chromosome 4). A footprint-occupancy score is informative only at Tn5 cut-site resolution (green, AUROC 0.68); the same score on smoothed coverage is inverted and uninformative (red, 0.24). Raw accessibility (grey) is near chance and the CTCF motif (blue) is the single strongest predictor.

4. Discussion

`fp-tools` contributes both a more accurate TF-binding predictor and a reproducible platform on which to run and benchmark it. On a genome-wide, chromosome-held-out benchmark its integrated score matches or exceeds the best individual paradigm in all 16 cell–TF tasks, with the largest margins precisely where footprints and motifs are individually weak—consistent with the long-standing observation that footprint depth is governed by factor residence time and local DNA sequence rather than occupancy alone [7], so a footprint-only or motif-only predictor is expected to fail for an important class of factors. Beyond accuracy, its comparative strengths are interface consistency across the classical footprinting workflow, deterministic regression coverage, and a conservative extension model in which new science is opt-in and non-disruptive (Table 3). These properties target precisely the friction that slows applied ATAC-seq analysis and cross-method benchmarking today.

Table 3. Feature comparison across widely used ATAC-seq regulatory-analysis tools. ✓, native first-class support; ○, partial or indirect support; —, absent. The `fp-tools` column covers released and development-branch functionality.

Feature	fp-tools	TOBIAS	HINT-ATAC	PRINT/ scPrinter	Chrom- BPNet	maxATAC
Bulk ATAC footprinting	✓	✓	✓	✓	○	—
Tn5 bias correction	✓	✓	✓	✓	✓	—
Classical footprint scoring	✓	✓	✓	—	—	—
Multiscale / nucleosome-aware	✓	—	○	✓	○	—
Supervised TFBS prediction	✓	—	—	✓	✓	✓
Variant scoring	✓	—	—	—	✓	✓
Motif-relaxed / motif-free recovery	✓	—	—	—	○	—
De novo motif discovery	✓	—	—	✓	✓	—
scATAC / pseudobulk support	○	○	○	✓	○	○
Visualization / reporting	✓	✓	○	✓	✓	○
GUI / YAML / batch execution	✓	—	—	—	—	—

We do not claim parity with deep sequence-to-profile models on every task where learned base-resolution sequence features dominate; ChromBPNet-class models and their attribution-based interpreters are powerful for sequence-driven occupancy and variant-effect prediction [6]. Those models, however, are computationally heavy—requiring GPU training and large compute budgets—and their learned enzyme-bias correction is itself incomplete and factor-dependent, whereas the `fp-tools` integrated predictor is a lightweight, interpretable, CPU-only model that is trained and applied in seconds and remains competitive on the tasks evaluated here. `fp-tools` makes classical and tabular-supervised analyses reproducible, composable, and easy to benchmark on shared labels. The multiscale, competition-decomposition, and replicate-uncertainty modules are pragmatic additions grounded in the available signal: in particular, the replicate-aware standard error is reconstructed from reported p-values rather than from per-replicate scores, so it should be read as a calibrated uncertainty heuristic rather than a full hierarchical model, and per-replicate inputs would enable a stronger formulation in future work.

Limitations remain. The present benchmark evaluates discrimination within each cell–TF task under chromosome-held-out cross-validation; whole-genome cross-cell-line transfer (training on some cell lines and testing on others) and direct head-to-head runs of external tools such as TOBIAS, HINT-ATAC, and ChromBPNet on the identical locus sets are the next steps before broad, field-relative superiority claims. Larger public-benchmark inputs depend on external data acquisition, intentionally kept outside version control for size and licensing reasons; the tabular supervised models trade base-resolution sequence modeling for transparency and speed; and several roadmap items (attribution-based discovery, RNA-prior weighting for pseudobulk, hierarchical differential binding) are scaffolded but not yet fully realized. These are natural directions for subsequent releases.

5. Conclusions

We presented `fp-tools`, a standalone, reproducible command-line platform that unifies classical ATAC-seq footprinting with opt-in multiscale, supervised, motif-discovery, variant-scoring, pseudobulk, replicate-uncertainty, and competition-decomposition modules. By keeping core workflows stable and test-covered while adding new scientific behavior as composable, independently validated modules—and by pairing them with a tiered public-data benchmark and figure infrastructure—the package lowers the barrier to standardized, benchmarkable regulatory-genomics analysis. A genome-wide, chromosome-held-out benchmark across four ENCODE cell lines and 16 cell–TF tasks shows that the integrated score matches or exceeds the best single predictor in every task and that optional isotonic recalibration yields well-calibrated probabilities. Future work will add whole-genome cross-cell-line transfer and direct external-tool comparisons, strengthen the supervised and hierarchical differential-binding models, and integrate attribution-based discovery.

Author Contributions: Conceptualization, Y.L. and C.Y.; methodology, Y.L.; software, Y.L.; validation, Y.L.; formal analysis, Y.L.; investigation, Y.L.; data curation, Y.L.; writing—original draft preparation, Y.L.; writing—review and editing, Y.L. and C.Y.; visualization, Y.L.; supervision, C.Y.; project administration, C.Y.; funding acquisition, C.Y. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable. This study used only publicly available, de-identified datasets and did not involve new experiments on humans or animals.

Informed Consent Statement: Not applicable.

Data Availability Statement: `fp-tools` is openly available at https://github.com/oncologylab/fp-tools. All benchmark manifests, schemas, analysis scripts, and figure generators are included in the repository. Public input datasets are obtained from the ENCODE Project [12], the 10x Genomics PBMC Multiome example [11], and motif databases (e.g., JASPAR [13]) via the included discovery and download scripts. Manifests, scripts, small result tables, source tables, and figures are stored in the main repository; large reusable example-data files are distributed separately through the public `oncologylab/fp-tools-data` repository release assets, and full raw public inputs are regenerated from committed manifests rather than stored in version control.

Acknowledgments: The author thanks the ENCODE Project Consortium and the maintainers of the open-source tools referenced in this work for making their data and software publicly available.

Conflicts of Interest: The author declares no conflicts of interest.

1. Buenrostro, J.D.; Giresi, P.G.; Zaba, L.C.; Chang, H.Y.; Greenleaf, W.J. Transposition of native chromatin for fast and sensitive epigenomic profiling of open chromatin, DNA-binding proteins and nucleosome position. *Nature Methods* **2013**, *10*, 1213–1218. https://doi.org/10.1038/nmeth.2688.
2. Bentsen, M.; Goymann, P.; Schultheis, H.; Klee, K.; Petrova, A.; Wiegandt, R.; Fust, A.; Preussner, J.; Kuenne, C.; Braun, T.; et al. ATAC-seq footprinting unravels kinetics of transcription factor binding during zygotic genome activation. *Nature Communications* **2020**, *11*, 4267. https://doi.org/10.1038/s41467-020-18035-1.
3. Li, Z.; Schulz, M.H.; Look, T.; Begemann, M.; Zenke, M.; Costa, I.G. Identification of transcription factor binding sites using ATAC-seq. *Genome Biology* **2019**, *20*, 45. https://doi.org/10.1186/s13059-019-1642-2.

4. Hu, Y.; Horlbeck, M.A.; Zhang, R.; Ma, S.; Shrestha, R.; Kartha, V.K.; Duarte, F.M.; Hock, C.; Savage, R.E.; Labade, A.; et al. Multiscale footprints reveal the organization of cis-regulatory elements. *Nature* **2025**, *638*, 779–786. https://doi.org/10.1038/s41586-024-08443-4.
5. Cazares, T.A.; Rizvi, F.W.; Iyer, B.; Chen, X.; Kotliar, M.; Bejjani, A.T.; Wayman, J.A.; Donmez, O.; Wronowski, B.; Parameswaran, S.; et al. maxATAC: Genome-scale transcription-factor binding prediction from ATAC-seq with deep neural networks. *PLoS Computational Biology* **2023**, *19*, e1010863. https://doi.org/10.1371/journal.pcbi.1010863.
6. Pampari, A.; Shcherbina, A.; Kvon, E.Z.; Kosicki, M.; Nair, S.; Kundu, S.; Kathiria, A.S.; Risca, V.I.; Hundenborn, K.; Zhou, Y.; et al. ChromBPNet: bias factorized, base-resolution deep learning models of chromatin accessibility reveal cis-regulatory sequence syntax, transcription factor footprints and regulatory variants. *bioRxiv* **2025**. Preprint, https://doi.org/10.1101/2024.12.25.630221.
7. Sung, M.H.; Guertin, M.J.; Baek, S.; Hager, G.L. DNase footprint signatures are dictated by factor dynamics and DNA sequence. *Molecular Cell* **2014**, *56*, 275–285. https://doi.org/10.1016/j.molcel.2014.08.016.
8. Pedregosa, F.; Varoquaux, G.; Gramfort, A.; Michel, V.; Thirion, B.; Grisel, O.; Blondel, M.; Prettenhofer, P.; Weiss, R.; Dubourg, V.; et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* **2011**, *12*, 2825–2830.
9. Bailey, T.L.; Johnson, J.; Grant, C.E.; Noble, W.S. The MEME Suite. *Nucleic Acids Research* **2015**, *43*, W39–W49. https://doi.org/10.1093/nar/gkv416.
10. Gupta, S.; Stamatoyannopoulos, J.A.; Bailey, T.L.; Noble, W.S. Quantifying similarity between motifs. *Genome Biology* **2007**, *8*, R24. https://doi.org/10.1186/gb-2007-8-2-r24.
11. 10x Genomics. PBMCs from a healthy donor (granulocyte sorted), single cell multiome ATAC + gene expression. 10x Genomics public dataset, Cell Ranger ARC 2.0.0, 2021. ATAC fragments, peak calls, TF-analysis tables, and gene-expression clustering outputs downloaded from 10x Genomics sample files.
12. The ENCODE Project Consortium. An integrated encyclopedia of DNA elements in the human genome. *Nature* **2012**, *489*, 57–74. https://doi.org/10.1038/nature11247.
13. Baydar, D.O.; Rauluseviciute, I.; Aronsen, D.R.; et al. JASPAR 2026: expansion of transcription factor binding profiles and integration of deep learning models. *Nucleic Acids Research* **2026**, *54*, D184–D193. https://doi.org/10.1093/nar/gkaf1209.